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Contribution

Model, Visualizations, Appendix Proofs
Likelihood Analysis, Abstract, Introduction,
Bayesian Analysis, Conclusion

Abstract

Write your abstract here.

Keywords

Likelihood Analysis, Bayesian Statistics, Frequentist Statistics, Vaccine Efficacy

Introduction & Background

Statistical Methods

Model

Describe the statistical model used.

Likelihood Inference

We are interested in ψ , the efficacy of the vaccine.

Using the distribution of T detailed above, this is its probability distribution function, where π is the probability of each of the 170 covid cases being from the vaccine group:

$$f(t) = \binom{170}{t} \pi^t (1 - \pi)^{170-t}, \quad t = 0, 1, \dots, 170$$

Using the proof in the appendix, we can write π in terms of vaccine efficacy ψ . $\pi = \frac{1-\psi}{2-\psi}$.

$$f(t) = \binom{170}{t} \left(\frac{1-\psi}{2-\psi}\right)^t \left(\frac{1}{2-\psi}\right)^{170-t}, \quad t = 0, 1, \dots, 170$$

The generalized likelihood function of ψ is the following:

$$L(\psi) = \binom{170}{t} \left(\frac{1-\psi}{2-\psi}\right)^t \left(\frac{1}{2-\psi}\right)^{170-t}$$

Now, the log-likelihood function:

$$\ell(\psi) = \ln \binom{170}{t} + t \ln(1 - \psi) - 170 \ln(2 - \psi) + (170 - t) \ln 1$$

$\hat{\psi}_0^{mle}$ will be the value of ψ such that the first derivative of the log-likelihood function is 0, and where the second derivative is analyzed to confirm that it is the global maximum.

$$\frac{d}{d\psi}\ell(\psi) = -\frac{t}{1-\psi} + \frac{170}{2-\psi} = 0$$

$$\frac{170}{2-\psi} = \frac{t}{1-\psi}$$

$$170 - 170\psi = 2t - t\psi$$

$$154 = 162\psi$$

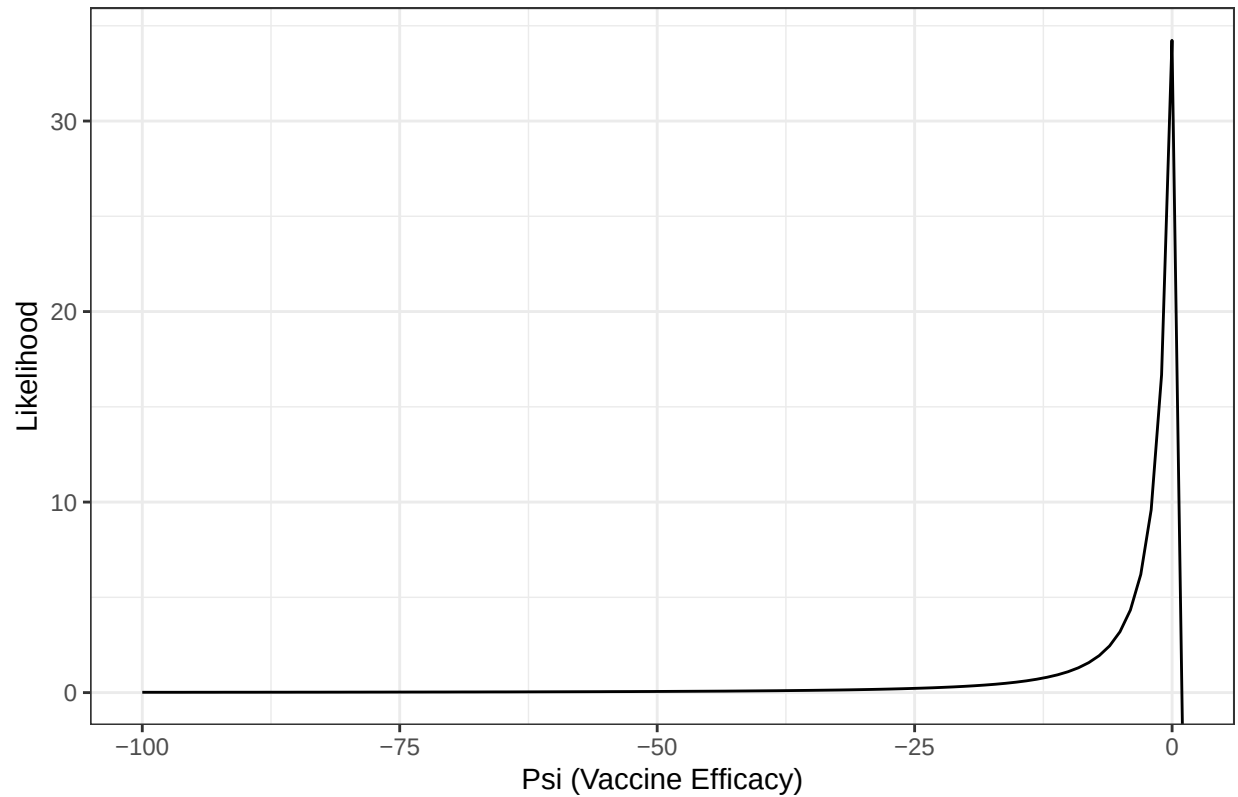
$$\hat{\psi}_0^{mle} = \frac{170 - 2t}{170 - t}$$

Since T is binomial, the maximum likelihood estimate of π is $\frac{t}{170}$. MLEs are invariant to transformation, so since $\psi = \frac{1-2\pi}{1-\pi}$, $\hat{\psi}_0^{mle} = \frac{1-2\frac{t}{170}}{1-\frac{t}{170}} = \frac{170-2t}{170-t}$, which is the same result as received with maximizing the likelihood function.

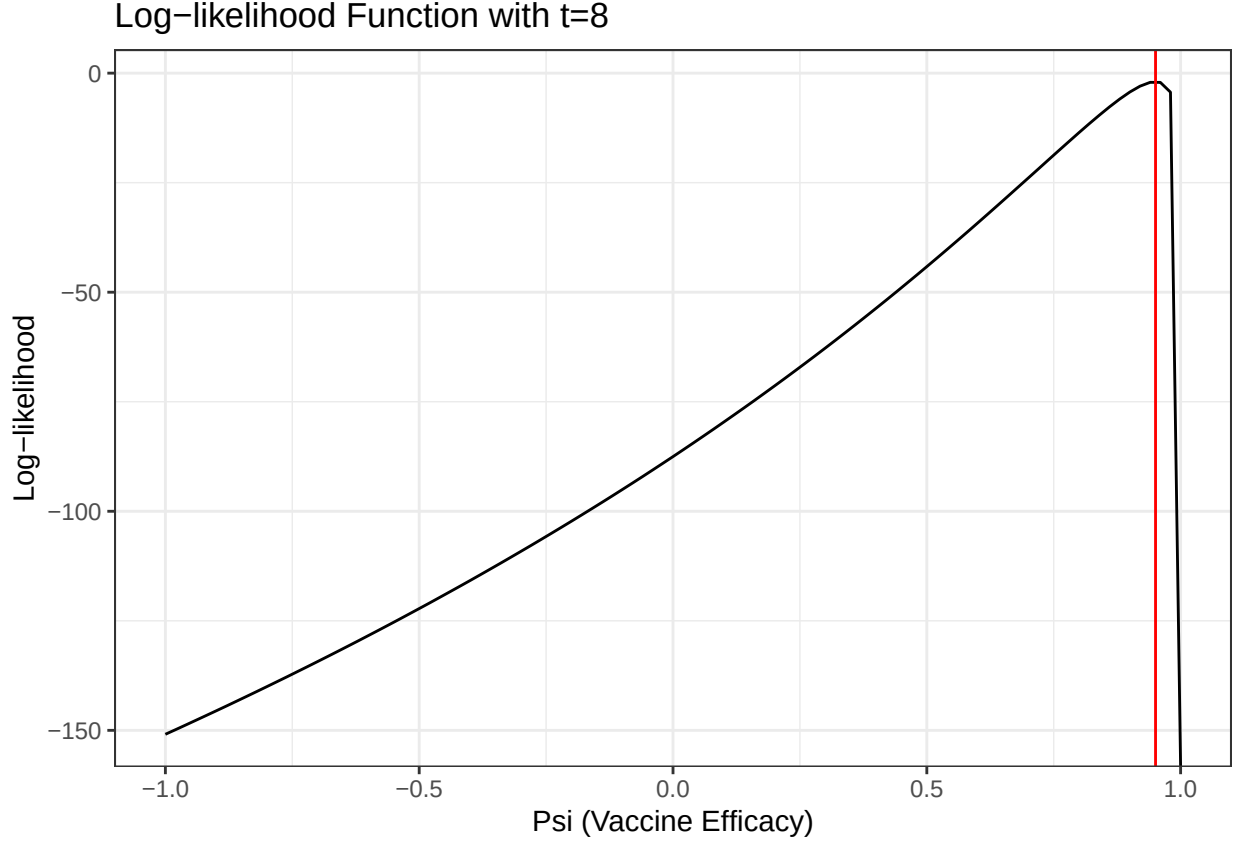
Second Derivative Test

$$\frac{d^2}{d\psi^2}\ell(\psi) = -\frac{t}{(1-\psi)^2} + \frac{170}{(2-\psi)^2}$$

Second Derivative of the Log-likelihood Function



The range of ψ is $-\infty < \psi < 1$. The second derivative of the log-likelihood function is positive for much of that range and is only negative between 0.723 and 1. This makes it difficult to confirm that 0.9506 is the global maximum likelihood estimate analytically. Instead, we'll look at a graph to confirm the maximum.



Between -1 and 1, the root of the first derivative has its maximum at 0.9506, our MLE. Below 0.9506, the first derivative is always positive, meaning that 0.9506 is the global maximum as well.

Confidence Intervals

A large sample 95% confidence interval was found for ψ_0 using the fact that for large samples and under certain regularity conditions, $\hat{\psi}_0^{mle}$ is approximately normal with mean ψ_0 and standard error $\frac{1}{\sqrt{I(\psi_0)}}$. With this approximate distribution, this confidence interval can be constructed:

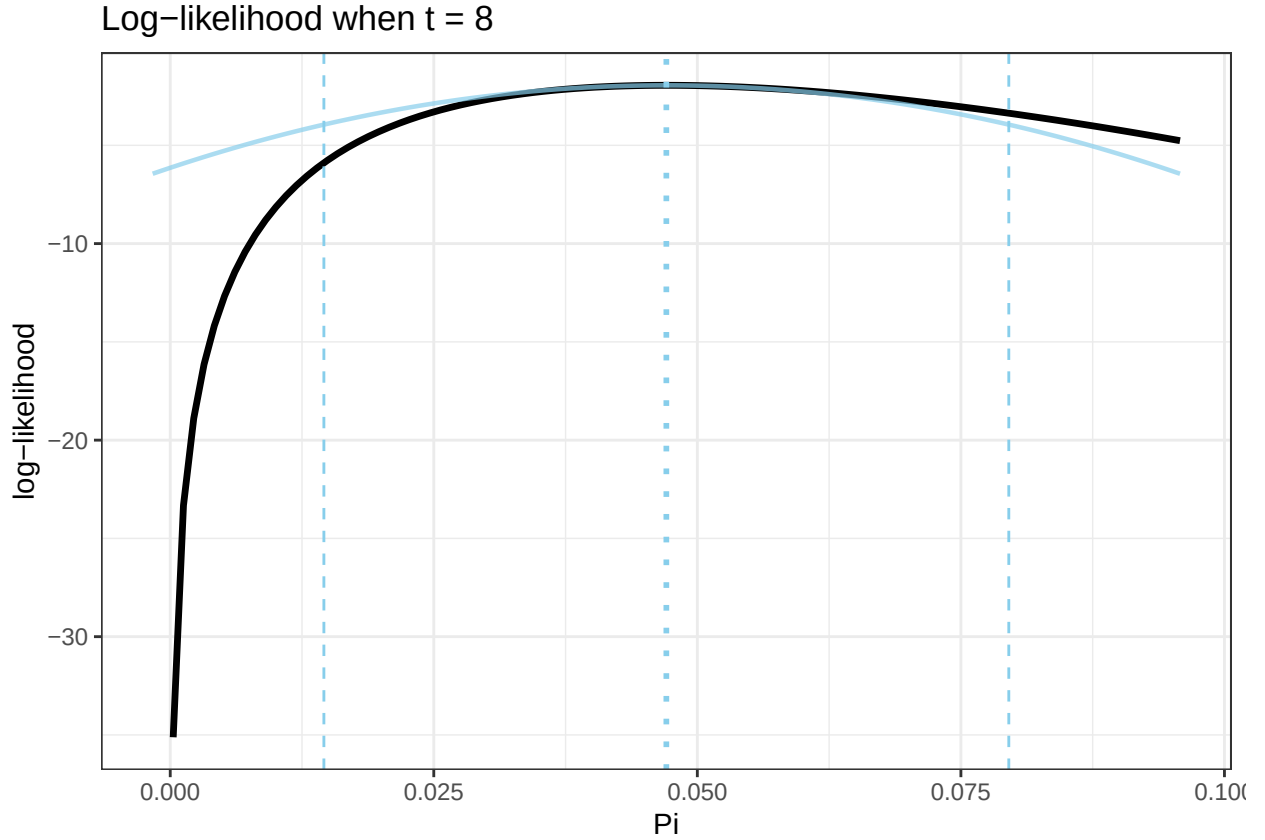
$$\hat{\psi}_0^{mle} \pm z_{0.025} \widehat{SE} = \hat{\psi}_0^{mle} \pm z_{0.025} \sqrt{\frac{1}{I(\psi_0)}}$$

$$I(\psi_0) = E\left[-\frac{d^2}{d\psi^2}\ell(\psi)\right]_{\psi=\psi_0} = E\left[\frac{t}{(1-\psi)^2} - \frac{170}{(2-\psi)^2}\right]_{\psi=\psi_0}$$

To achieve the expected Fisher information, we used our estimate for ψ_0 , so $I(\psi_0) \approx \frac{t}{(1-\hat{\psi}_0^{mle})^2} -$

$$\frac{170}{(2 - \hat{\psi}_0^{mle})^2}.$$

To assess the ease of finding $\hat{\psi}_0^{mle}$ numerically, here is the log-likelihood function when $t = 8$ next to its second order Taylor series. Because of numerical difficulties, the function is plotted in terms of π , not ψ , but the convergence of the two functions is the same as what it would be in terms of ψ .



Between the blue lines, the log-likelihood function matches the second order Taylor expansion well. This means that the log-likelihood function is close enough to quadratic that the second order Taylor series is a good approximation of it. If the second order Taylor series wasn't a good fit for it, there would be risk of an inaccurate estimate.

They diverge farther from the MLE, because the log-likelihood function is not polynomial and it's only the second order Taylor approximation, so the approximation does not last.

We can check the estimate given by Newton-Raphson, $\hat{\pi}_0^{mle} = 0.04706$, by remembering that π is the probability parameter of a binomial distribution and finding MLE $\frac{t}{170} = \frac{8}{170} \approx 0.04706$, which matches the Newton-Raphson estimate.

Bootstrapping

Now, to find a confidence interval for the estimate, we'll take 10000 bootstrapped estimates of the MLE, and These values were generated from the distribution of T with π determined by our estimate of the vaccine efficacy, 0.9506.

We'll then take the 2.5th and 97.5th percentiles of the generated values to find the middle 95% of the set. Those numbers will be the bounds of the confidence interval for the vaccine efficacy ψ .

Likelihood ratio significance test

Then, we'll perform a significance test on the maximum likelihood estimate using the chi square distribution. This is a two-sided significance test, where the alternative hypothesis is that ψ is not 0.3.

Statistic $W = 2\ell(\hat{\psi}_0^{mle}) - 2\ell(\psi_{null})$ is approximately a chi square variable with 1 degree of freedom, so the chi square distribution is used to find a p-value for the data.

The p-value is very close to 0. By this value, we reject the null hypothesis in favor of the vaccine efficacy not being 0.3.

Empirical p value

For an empirical p-value, 10,000 values of T are drawn from the distribution associated with the null hypothesis: that the vaccine efficacy was 30%. This meant the trials of T each had a $\frac{1-0.3}{2-0.3} = \frac{7}{17}$ probability of being from the vaccine group.

The empirical p-value is the proportion of $W = 2\ell(\hat{\psi}^{mle}) - 2\ell(0.3)$ (where $\hat{\psi}^{mle}$ is the MLE based on the null-generated data) that exceed the value of W where $\hat{\psi}^{mle}$ is the MLE based on the study observations.

Bayesian Inference

Results

$$\hat{\psi}_0^{mle} = \frac{170 - 2t}{170 - t} = \frac{154}{162} \approx 0.9506$$

The maximum likelihood estimate for the efficacy of the BNT162b2 vaccine is 0.9506. This means that we estimate that the vaccine decreases your risk of a covid infection by 95.06%.

95% large sample confidence interval around the estimate is [0.9155627, 0.9856719], meaning that there is 95% confidence that the true efficacy is between 91.55% and 98.57%.

The 95% confidence interval using bootstrapping was [0.9102564, 0.9820359], meaning that there is 95% confidence that the true efficacy is between 91.03% and 98.20%.

The chi square derived p-value was 2.82e-28, which is extremely small. This means we reject

The empirical p-value was 0, meaning that in 10000 data points generated from the distribution with a vaccine efficacy of 0.3, none were more extreme than the study data. By this value, we reject the null hypothesis in favor of the vaccine efficacy not being 0.3. It is very strong evidence of a different vaccine efficacy.

In both methods for finding a p-value, the null hypothesis was rejected.

Discussion & Conclusion

Bibliography

Polack et al. (2020). Safety and Efficacy of the BNT162b2 mRNA Covid-19 Vaccine. *New England Journal of Medicine*, 383(27).

Grove, R. (2026). *Physician's Health Study*. Lecture slides, Introduction To Probability And Mathematical Statistics III, University of Washington.

Senn, S. (2022). The design and analysis of vaccine trials for COVID-19 for the purpose of estimating efficacy. *Pharmaceutical Statistics*, 21(4).

Appendix

Code

Likelihood

```
# functions
log_lik <- function(psi){
  log(choose(170,8))+8*log(1-psi)-8*log(2-psi)-162*log(2-psi)
}

second_deriv <- function(psi){
  -8/((1-psi)^2) + 170/((2-psi)^2)
}

ggplot() +
  geom_function(fun=second_deriv, xlim=c(-100,1)) +
  theme_bw() +
  labs(title="Second Derivative of the Log-likelihood Function") +
  xlab("Psi (Vaccine Efficacy)") +
  ylab("Likelihood")
```

```
# loglik graph
ggplot() +
  geom_function(fun=log_lik, xlim=c(-1,1)) +
  geom_vline(aes(xintercept=0.9506), color='red') +
  theme_bw() +
  labs(title="Log-likelihood Function with t=8") +
  xlab("Psi (Vaccine Efficacy)") +
  ylab("Log-likelihood")
```

```

# taylor series
log_lik_pi <- function(pi){
  log(choose(170,8))+8*log(pi)+162*log(1-pi)
}

ml <- maxLik2(loglik=log_lik_pi, start=0.5)

plot(ml) +
  theme_bw() +
  labs(title="Log-likelihood when t = 8") +
  xlab("Pi")

```

```

# confidence interval
data <- data.frame(Group=c("BNT162b2","Placebo","Total"), Cases=c(8,162,170), Sample_size=c(170,170,170))

t <- data[data$Group=="BNT162b2", "Cases"]
n <- data[data$Group=="Total", "Cases"]
mle <- (n-2*t)/(n-t)
se <- 1/sqrt(8/((1-mle)^2)-170/((2-mle)^2))
z <- qnorm(0.975)

lower <- mle - z * se
upper <- mle + z * se

cat("[",lower,", ",upper,"]",sep="")

```

```

# bootstrapping
set.seed(414)

B <- 10000

```

```

sample_size <- 15
pi <- (1-mle)/(2-mle)

boot_df <- as.data.frame(replicate(n=B, expr={x <- rbinom(n=1,size=170, prob=pi)
                                     (170 - 2*x)/(170-x)
                                     }))
names(boot_df) <- c("est")

lwr <- quantile(boot_df$est, 0.025)
upr <- quantile(boot_df$est, 0.975)

cat("[",lwr," ",upr,"]",sep="")

```

```

# chi p value
null <- 0.3
W <- 2*log_lik(mle)-2*log_lik(null)
pchisq(W, df=1, lower.tail=F)

```

```

# emp p value
set.seed(414)

unifs <- replicate(n=B, expr={x <- rbinom(n=1, size=170, prob=(0.7/1.7))
                             mle <- (170-2*x)/(170-x)
                             W <- 2*log_lik(mle)-2*log_lik(null)
                             })

pvalue <- sum(unifs>W, na.rm=T)/B
cat("Proportion Exceeding W:", pvalue)

```

Proofs

Derivation of log-likelihood function in terms of pi

$$f(t) = \binom{170}{t} \pi^t (1 - \pi)^{170-t}, \quad t = 0, 1, \dots, 170$$

$$L(\pi) = \binom{170}{8} \pi^8 (1 - \pi)^{162}$$

$$\ell(\pi) = \ln \binom{170}{8} + 8 \ln \pi + 162 \ln(1 - \pi)$$